



राष्ट्रीय प्रौद्योगिकी संस्थान पटना

National Institute of Technology Patna

Mid Semester Examination (Jan-Jun'22)

Session: 2021-22 Spring'22 Semester

Department: Computer Science and Engineering

Programme: B.Tech.-CSE

Semester: 2nd

Course Code: **CS24108**

Course: **Computer Organization**

Full Marks: 30

Duration: 2 hours

[Attempt all questions; Answer concisely only in blue/black ink; Use pencil for artwork;
Assume missing data; Any form of calculator is **not permitted** for this course]
[Marks, Course Outcome and Bloom's Level are mentioned on right-hand side of each question]

Sl. No.	Questions	CO	BL
1. a.	Write flowcharts of restoring division and non-restoring division methods. (3+3) [Course outcome(s) evaluated: CO-1(Remember/Recall)]	CO-1	Level-1
b.	Multiply the following given two's complement binary integers by Booth's multiplication algorithm. Show the intermediate steps in tabular form. (4) Given: Multiplicand (M)= $5_{10}=0101_2$, Multiplier (Q)= $-6_{10}=1010_2$. [Course outcome(s) evaluated: CO-3(Apply/Solve)]	CO-3	Level-3
2. a.	Differentiate between single-precision (i.e. 32-bit) and double-precision (64-bit) formats of IEEE 754 floating-point standard using schematic diagram. (4) [Course outcome(s) evaluated: CO-1(Remember/Recall)]	CO-1	Level-1
b.	Illustrate the algorithm of addition of IEEE 754 floating-point numbers using schematic diagram. (4) [Course outcome(s) evaluated: CO-1(Remember/Recall)]	CO-1	Level-1
c.	How to align the significands of addends $A = 1.001110_2 \times 2^3$ and $B = 1.001_2 \times 2^{-1}$ before performing their addition? (2) [Course outcome(s) evaluated: CO-3(Apply/Solve)]	CO-3	Level-3
3. a.	Consider the adjacent data path diagram. Consider a (macro)instruction: ADD R0, R1, R2 (through which $R0 \leftarrow R1 + R2$ is performed). Consider that ADD instruction is executed over the given data path, while assume that PC is incremented appropriately. Explain the sequence of microoperations (in terms of data path components) required to execute ADD instruction, if instruction cycle comprises of Fetch, Decode and Execute phases? (9) [Course outcome(s) evaluated: CO-2(Understand/Explain)]	CO-2	Level-2

